Advanced Driver Assistance Systems (ADAS)

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Agenda

- Introduction to ADAS
- ADAS sensors
- ADAS features

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Automotive Features

- **Set 1**
  - Comfort
  - Convenience
  - Safety
  - Security

- **Set 2**
  - Driver Assistance
  - Automaticity
  - Connected
  - Shared

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Introduction to ADAS



Source: https://www.loc.gov/pictures/collection/ffcarp/item/99614816/



Source: https://www.timetoast.com/timelines/evolution-of-steam-power-during-the-industrial-revolution



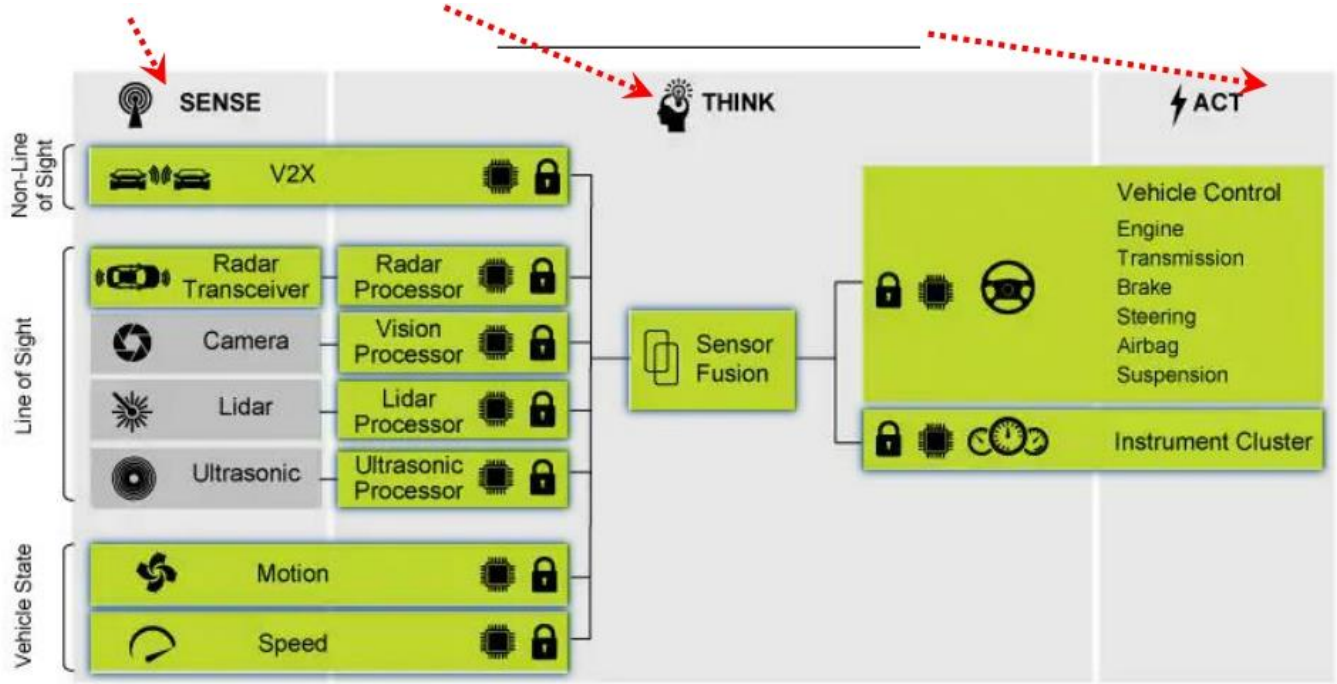
Source: https://www.timetoast.com/timelines/evolution-of-steam-power-during-the-industrial-revolution

- **Steam power revolutionized 18th century England and brought about the Industrial Revolution around the world**
- **Digital technology will enable the change in 21st century and create the AI Revolution**

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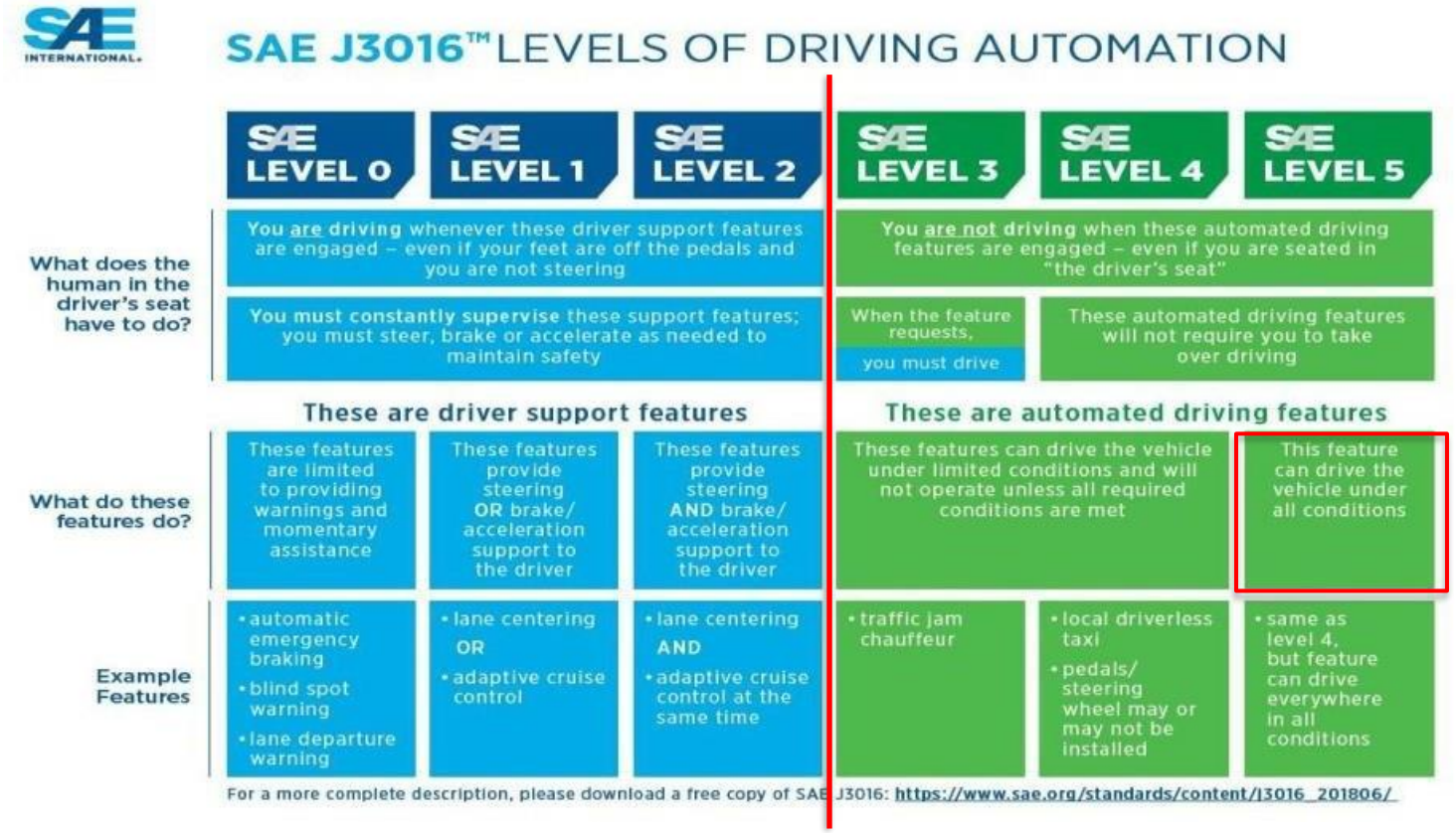
Introduction to ADAS

A **multi sensor network** and **decision engine** capable of processing high volumes of data in real-time to take **safety critical vehicle actions**



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Introduction to Automation Levels



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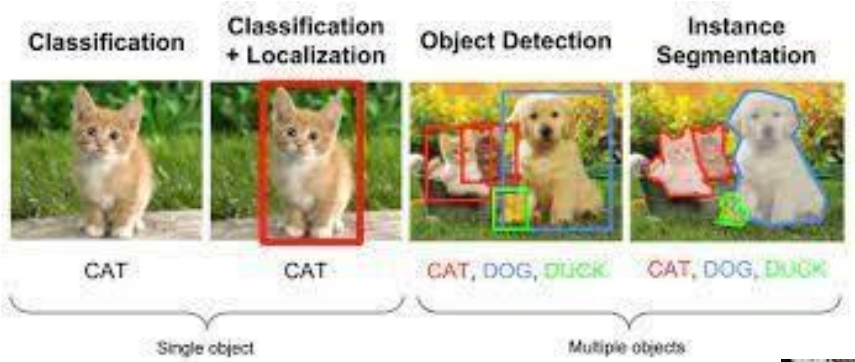
Localization

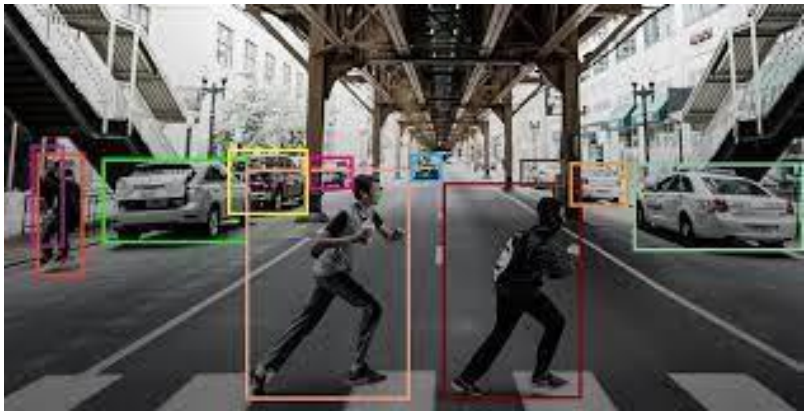
- Localization is how a car figures out where it is in the world.
- Previously, we discussed about computer vision, which is how cars
  use cameras to understand their environments.
- Then we will look at sensor fusion, which is how cars combine data from other sensors, such as lidar and radar, to augment their perception.



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Perception & Localization





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Take Next Logical Action

PERCEPTION	LOCALIZATION	PATH PLANNING	CONTROL

COMPUTER VISION

Detection of roads, signs, lights, objects, …

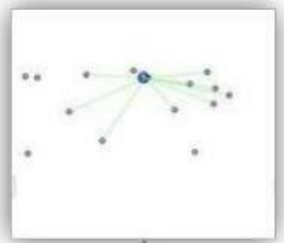


SENSOR FUSION

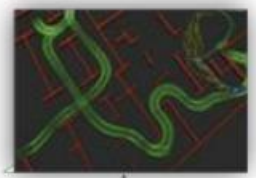
Detection of vehicles, obstacles, objects…



Understanding the world



Localization & Mapping : « **Where are we ?** »



Calculating trajectories from our position and our understanding of the world



Following the calculated trajectory

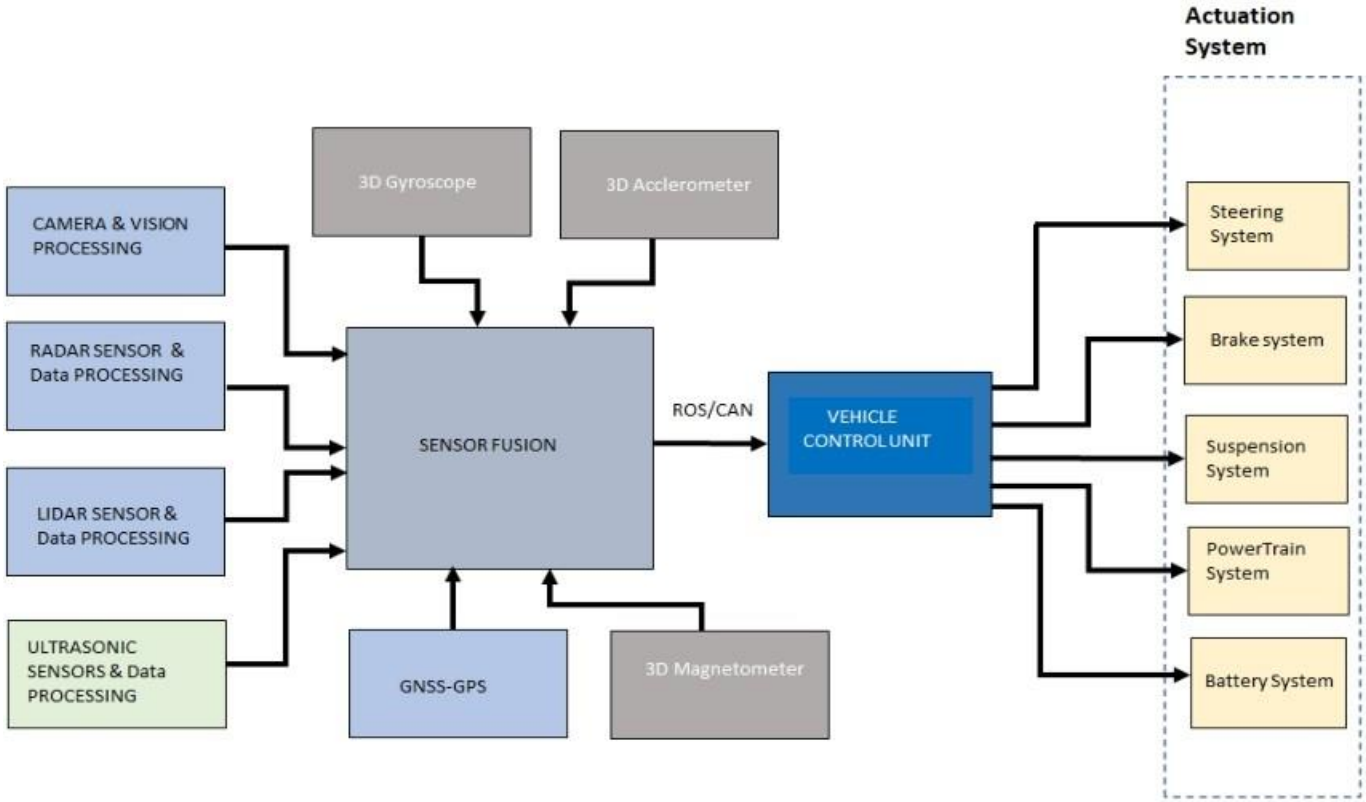
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Sensors/Technologies

- LiDAR (Light Detection and Ranging)
- RADAR (Radio Detection and Ranging)
- CAMERA
- Ultrasonic Sensor
- IMU (Inertial Measurement Unit)
- GNSS/GPS (Global Navigation Satellite System/Global Positioning System)

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Block Diagram



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Audio Visual (AV) Components

Sensors:

- Connected to networks or the cloud, sensors yield real-time, actionable data that can power AV
- Most of these devices have limited range and bandwidth and must be improved and fused with other sensor data to power a Level 5 Autonomous Driving Experience.

Processors:

- With the increasing requirements for processing speed in AV applications, processors are used for everything from building a real-time 3D spatial model of a car's surroundings to calculating proximity and threat levels based on the environment

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AV Components

Software:

- Automotive companies increasingly rely on software
- After all, the software makes the hardware work
- The hardware can be significantly simplified with software, especially when machine learning and artificial intelligence can be implemented to manage different situations
- The development effort behind software is huge in the industry

Mapping:

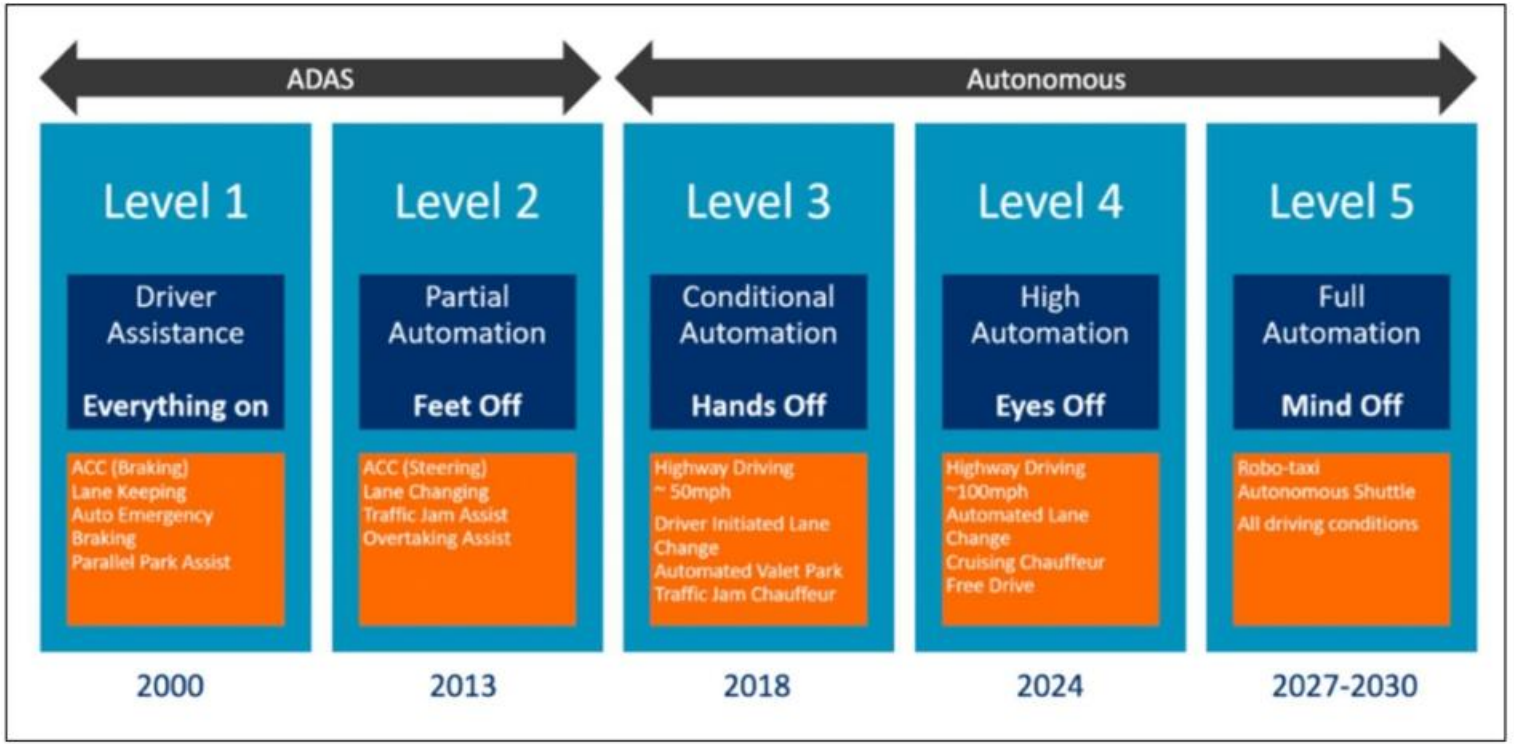
- The AV mapping function stores and updates geographical and infrastructure information gathered via vehicle sensors
- To determine its exact location
- This function maintains the information and communicates it to system control even if GPS fails

Actuators:

- The electrification of the actuator system has been a major enabler of AV because it has facilitated interaction with other electrical components in the vehicle
- With processors to analyze data from vehicle sensors, the AV system can make decisions executable by actuators

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Introduction to ADAS



Source: IEEE Spectrum

- **From an individual's perspective, this captures the level of involvement required**

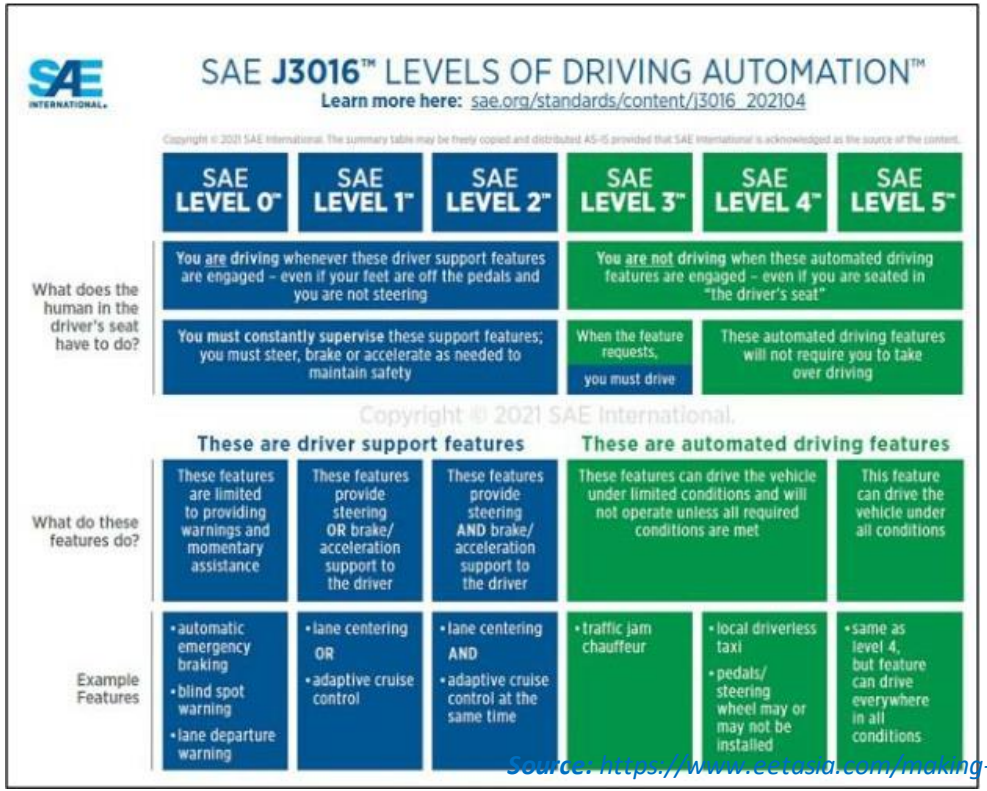
SAE (Levels of automation)

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Introduction to ADAS

SAE (Levels of automation)

human drivers



Replace human drivers

Source: https://www.eetasia.com/making-sense-of-the-vehicle-automation-levels

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Introduction to ADAS

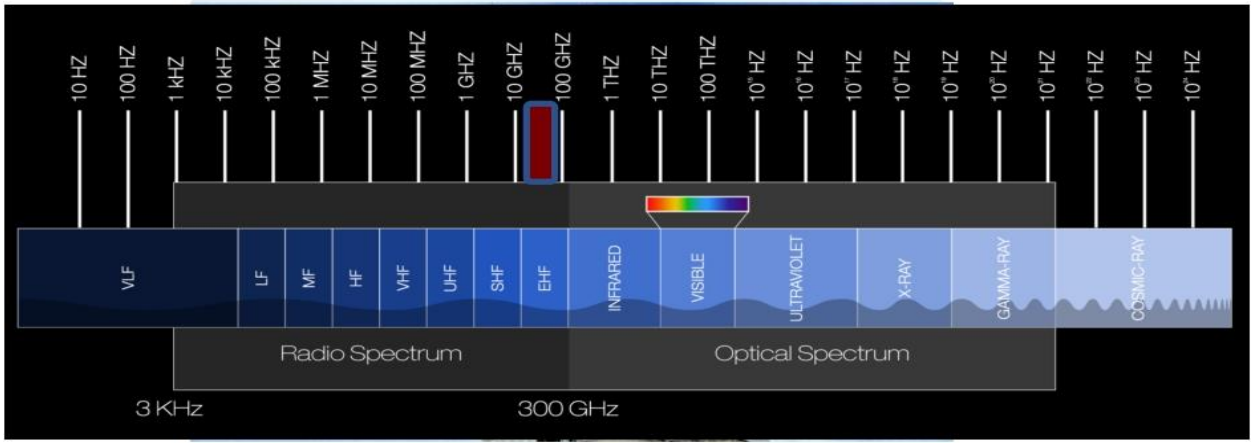


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ADAS Sensors

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ADAS Sensors - Radar



- Radar has been developed and used extensively in military, aerospace and law enforcement field
- Today it is moving towards industrial, automotive and even consumer applications
- Radio and Optical frequencies form the Electromagnetic spectrum; Radio spectrum is from 3kHz to 300GHz
- Automotive Radar operates in the K-band (18 – 27 GHz) and W-band (40 – 100+ GHz)

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ADAS Sensors - Radar



- From assistance systems to fully autonomous driving, Radars are featured in multiple use cases
- Long, Medium & Short range radar work on the same principle, but optimized for different detection capabilities

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ADAS Sensors - Radar



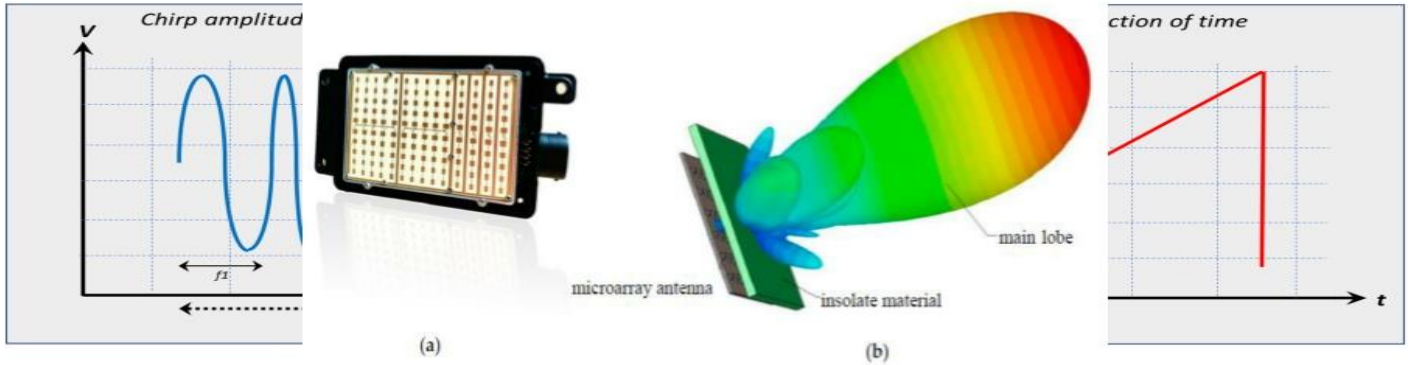
Source: Vayyar Imaging

In cabin monitoring using radar being developed for

- Occupancy detection
- Driver alertness sensing
- Airbag deployment

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ADAS Sensors - Radar



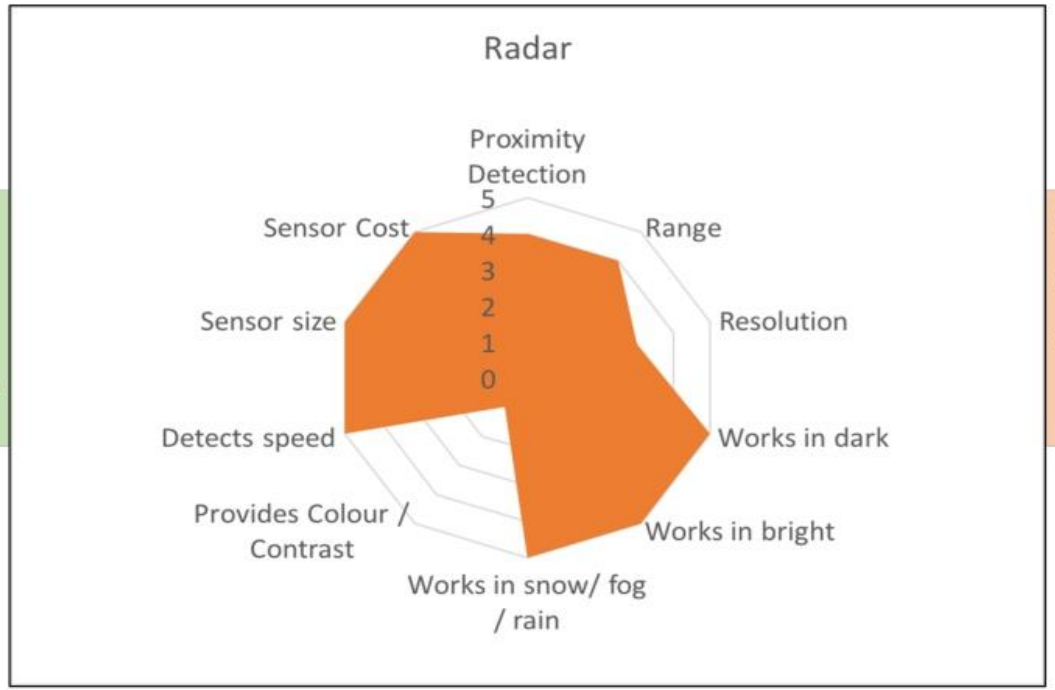
- Automotive radar works on the principle of measuring frequency shift between transmitted and received signals
- Radar is used to provide range, relative velocity and angle of approach information
- The transmit signal is called chirp, which is a sinusoid with frequency linearly increasing as function of time.

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ADAS Sensors - Radar

Pros:

- **Velocity detection**
- **Range**
- **All-light & weather condition**

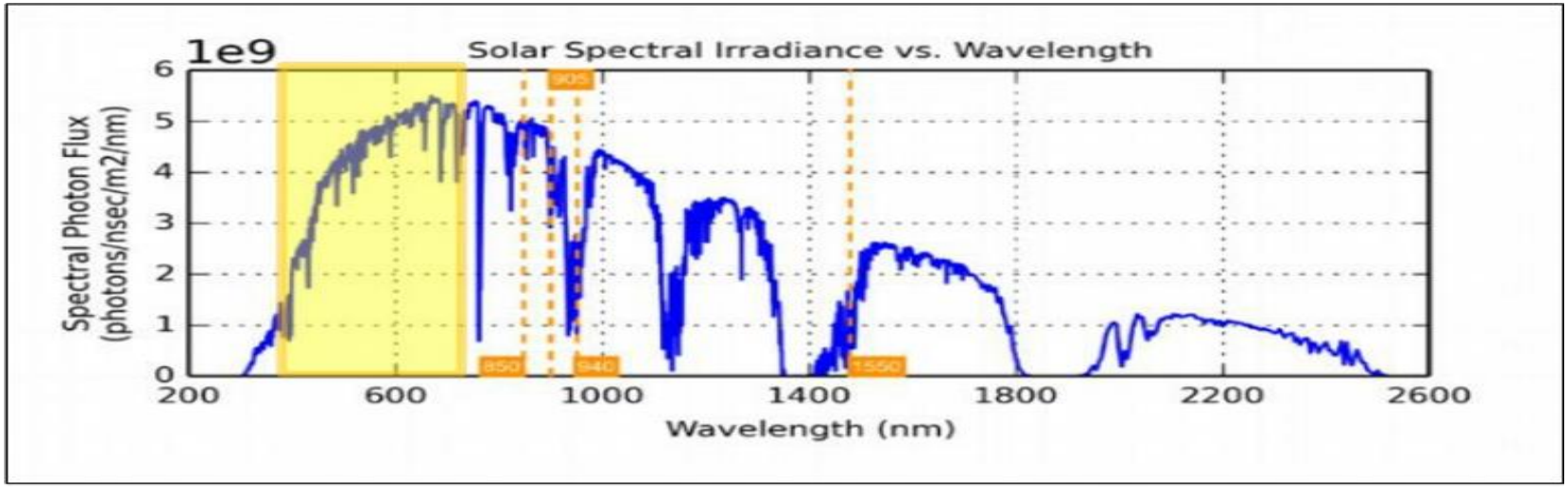


Cons:

- **Resolution**
- **Colour contrast**

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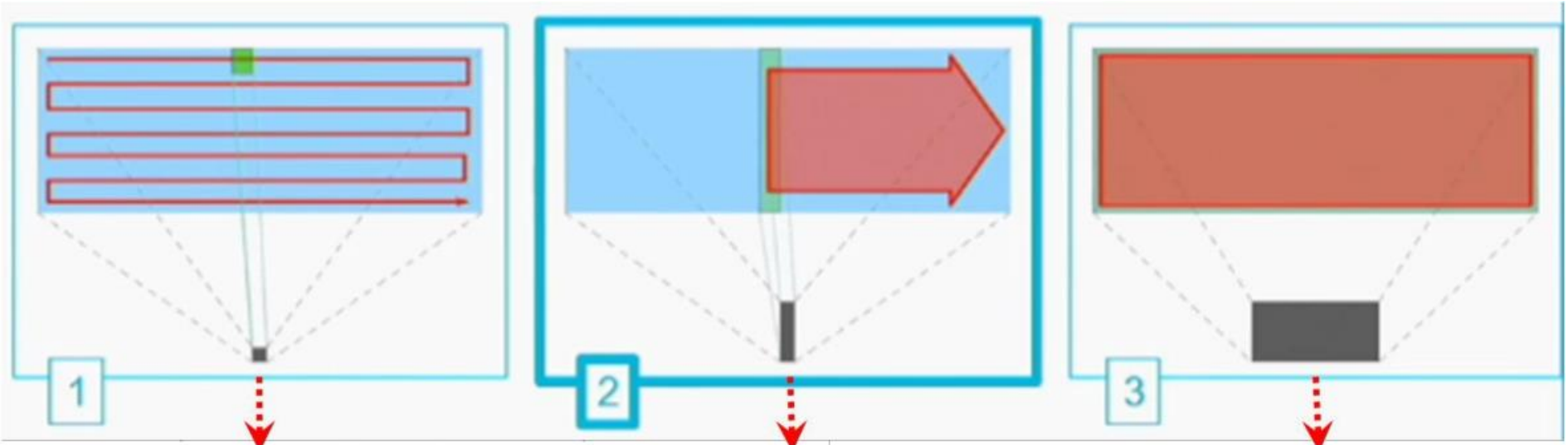
ADAS Sensors - Lidar



- Natural sunlight has nulls or low emission wavelengths (850, 900, 940, 1100, 1400nm)
- Laser light in the visible spectrum (400 – 700nm) cannot be transmitted due to driver distraction
- Light in infra-red (IR) frequency spectrum is used
  - Near IR (NIR) is 800 - 950nm
  - Far IR (FIR) is 1500 - 1600nm

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ADAS Sensors - Lidar



Raster (X-Y) Scan

Number of scans = 10000

Time for each scan = 10us

Line Scan

Number of scans = 200

Time for each scan = 500us

Flash Scan

Number of scans = 1

Time for each scan = 100ms

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ADAS Sensors - Lidar

No moving parts; single shot scene

Spinning



Velodyne
Uber

Flash



Leddar Tech
GM

MEMS micro-mirrors



Innoviz
BMW

Optical Phase Array



Quanergy
Nissan

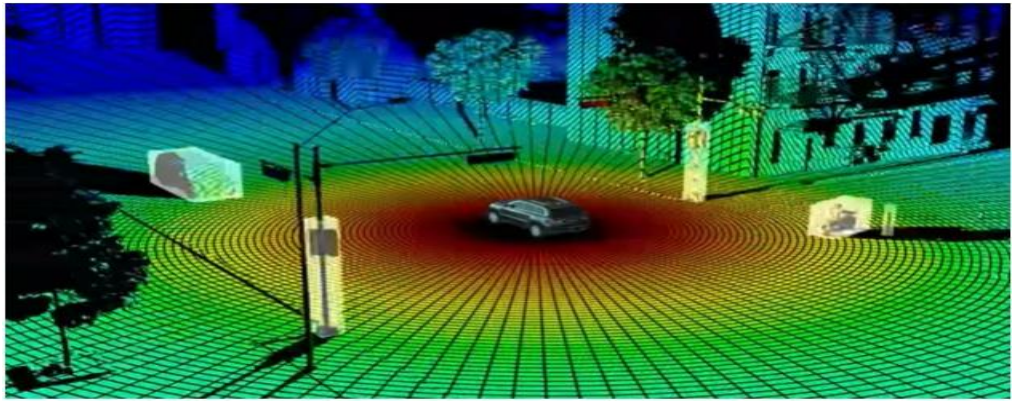
Mechanical rotation involved; most bulky and expensive

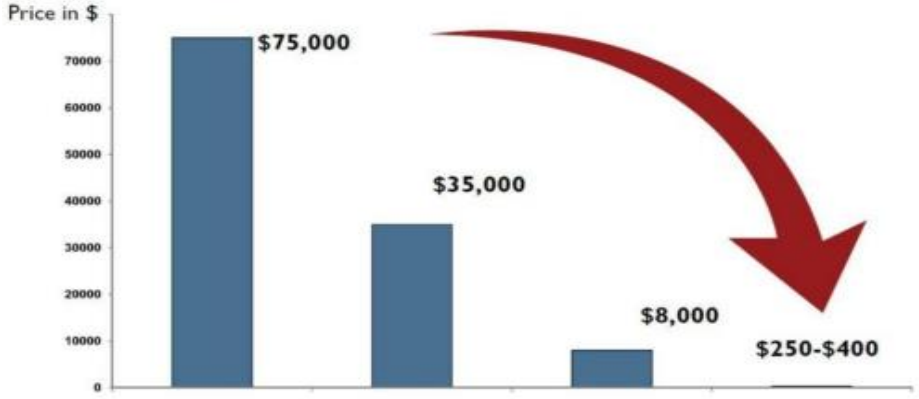
Most compact and cost-effective

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ADAS Sensors - Lidar

- Reduction in cost & size (solid-state Lidar)
- High reliability due to no moving parts
- Lower power requirements
- Higher resolution & refresh rates
- Better mechanical integration into body of the vehicle

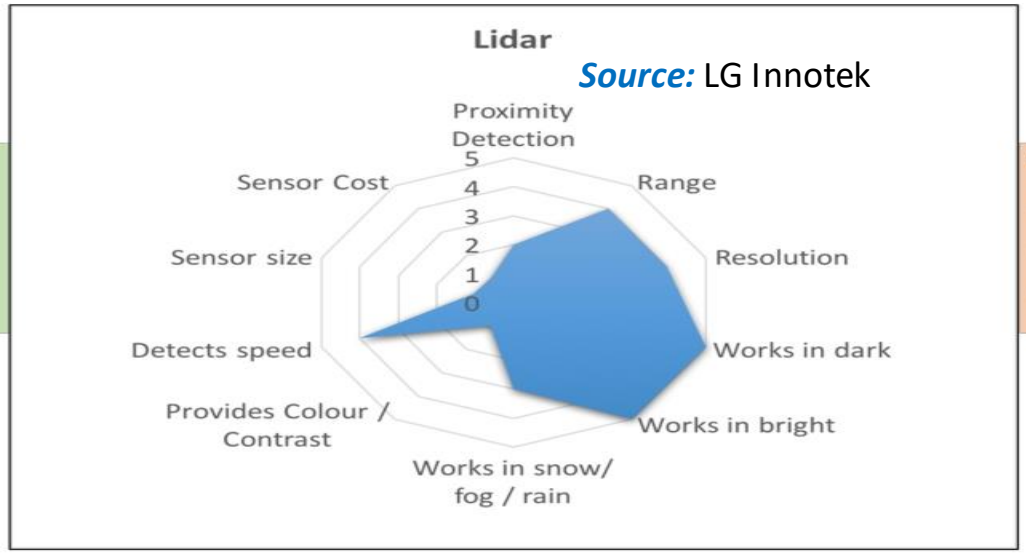




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ADAS Sensors - Lidar

Pros:
- **Resolution**
- **Range**
- **All-light condition**



Cons:
- **Affected by weather**
- **Size & Cost**
- **Colour contrast**

Source: Omnivision

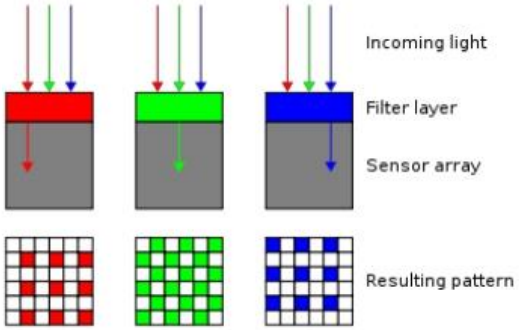
Source: https://cleantechnica.com/2016/07/29/tesla-google-disagree-lidar-right/

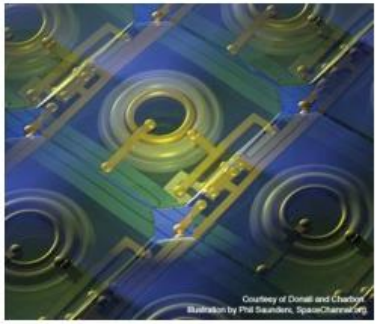
- Single photo avalanche detector (SPAD) for photo-detection; P-N diode in reverse-breakdown region for activation
- For colour capture, a Bayer (RGBG) filter (micro-lens arrangement) is used to capture primary colour palate

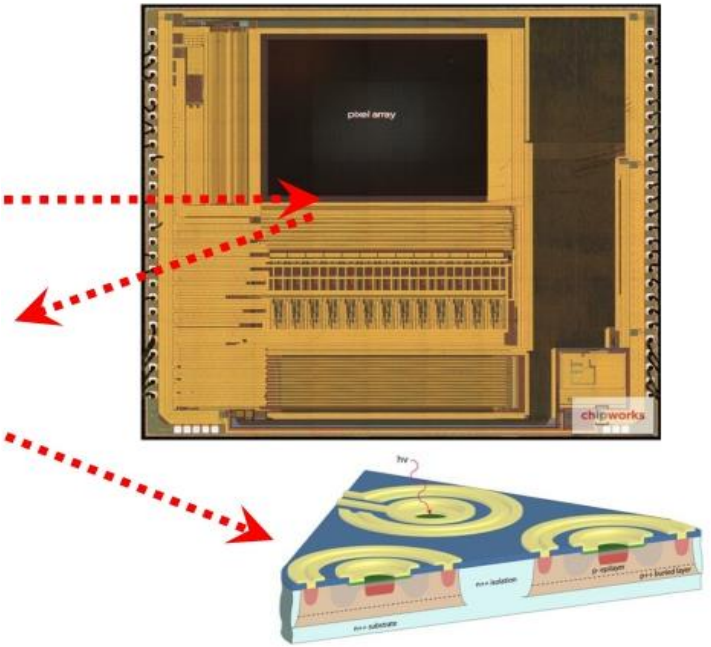
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ADAS Sensors - Camera



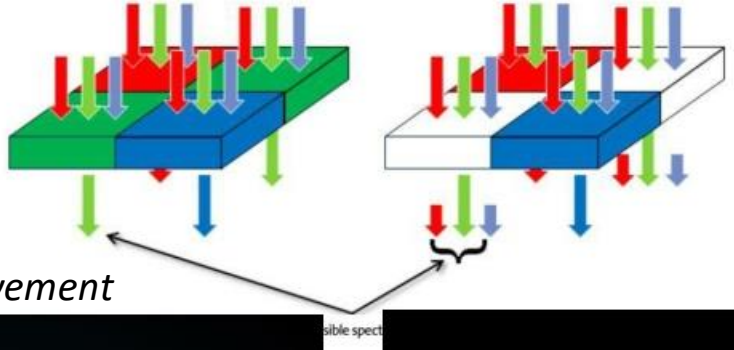






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ADAS Sensors - Camera



Low-light performance improvement

LED flicker mitigation range

High dynamic

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ADAS Sensors - Camera

- **Object detection and classification**
- Lane detection (Real and virtual)
- Traffic Sign Detection
- Traffic light classification
- Curb Detection
- Detection of speed bumps

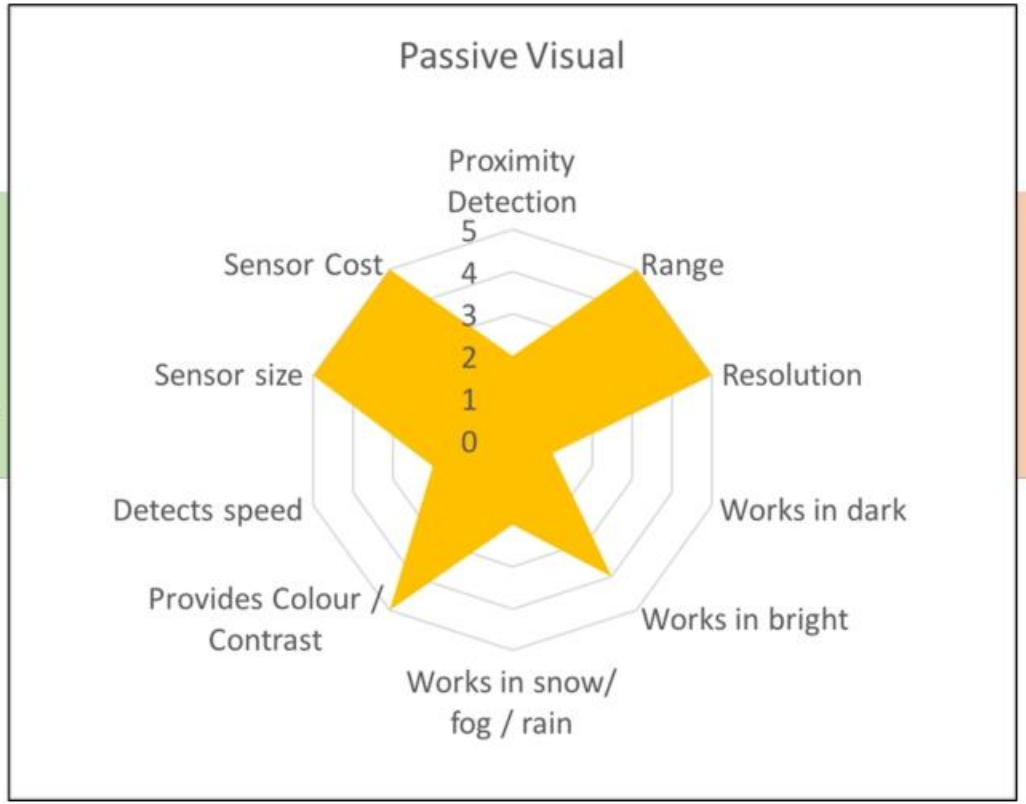


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ADAS Sensors - Camera

Pros:

- **Colour contrast**
- **Size & Cost**
- **Range & resolution**

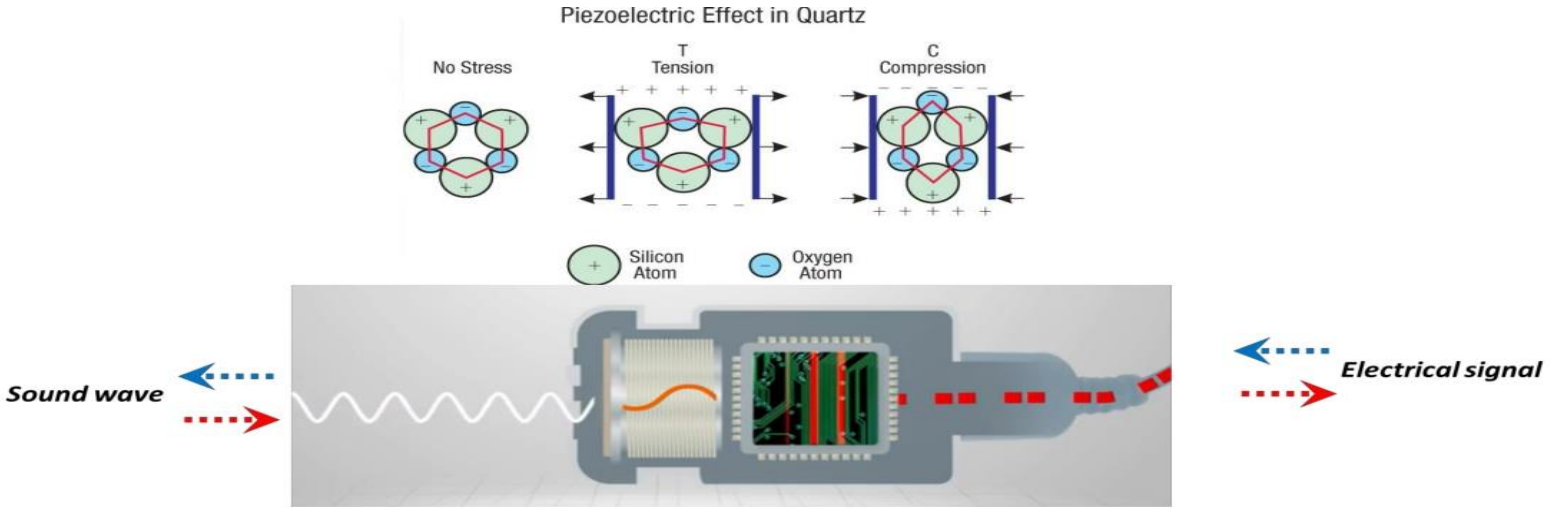


Cons:

- **Affected by weather**
- **Low-light conditions**
- **Velocity estimation**

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ADAS Sensors - Ultrasonic



- Under pressure, Quartz or PZT crystals create polarizing electric field (*microphone*)
- Alternatively, an electric field creates movement (*speaker*)

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Ultrasonic Sensors

Ultrasonic Sensors

- Uses sonic waves, in the range 20KHz to 58Khz,generated by a magneto resistive membrane, to measure distance of the object
- Its principle is based on the measurement of the time of flight(TOF) of the sonic wave from when it is emitted until the echo is received

$$d = \frac{c}{2} \times \text{ToF}$$

- The c velocity of the wave is in meters per second and TOF is the time of flight in seconds.
- Parking systems or short distance measurement at low speed.



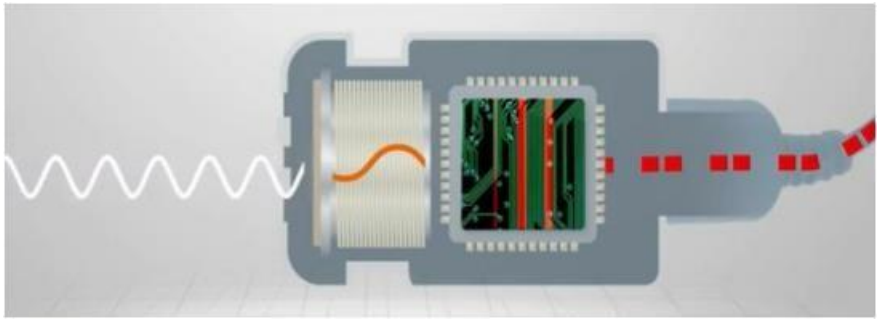


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Ultrasonic Sensor

- Under pressure, Quartz or PZT
- Crystals create polarizing electric field (*microphone*)

Sound wave

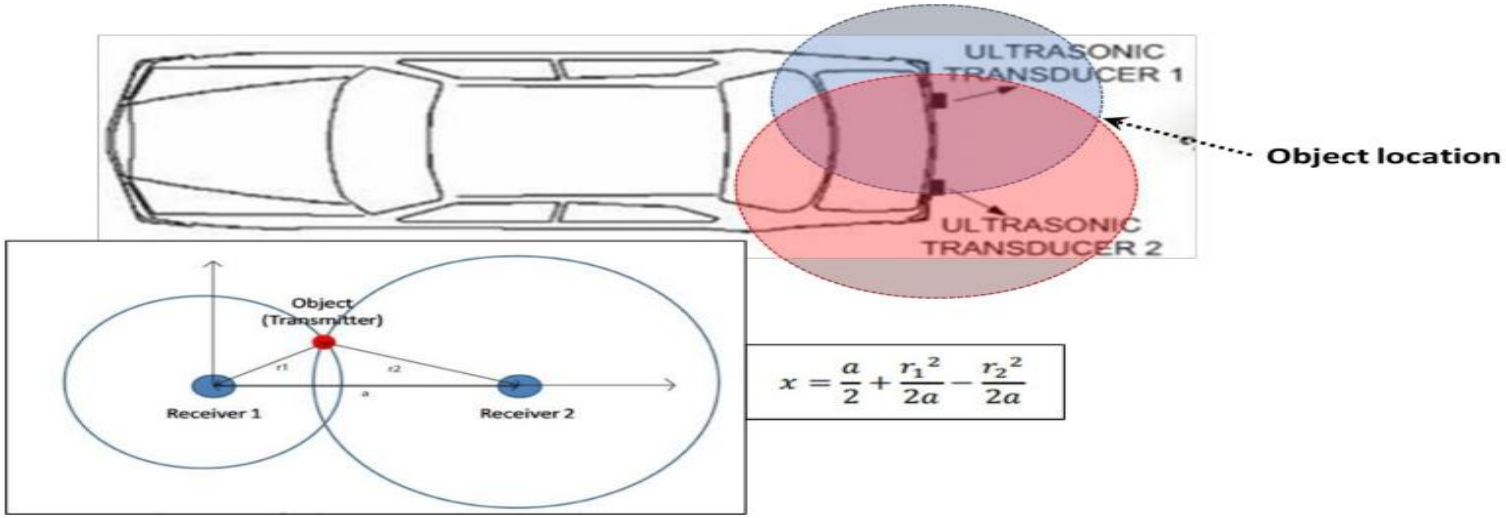


Electrical signal

Source: *AutoMate*

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ADAS Sensors - Ultrasonic



- Trilateration techniques used to identify objects and distance to the receiver
- One of the two solutions is inside the car; hence can be discarded (50% of area only needs to be surveyed)
- Three sets of sensors each in four corners (12 in all)
- Utilized for blind spot, park assist and lane changing

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ADAS Sensors - Ultrasonic



Source: https://www.youtube.com/watch?v=5F_mOOz3dV4

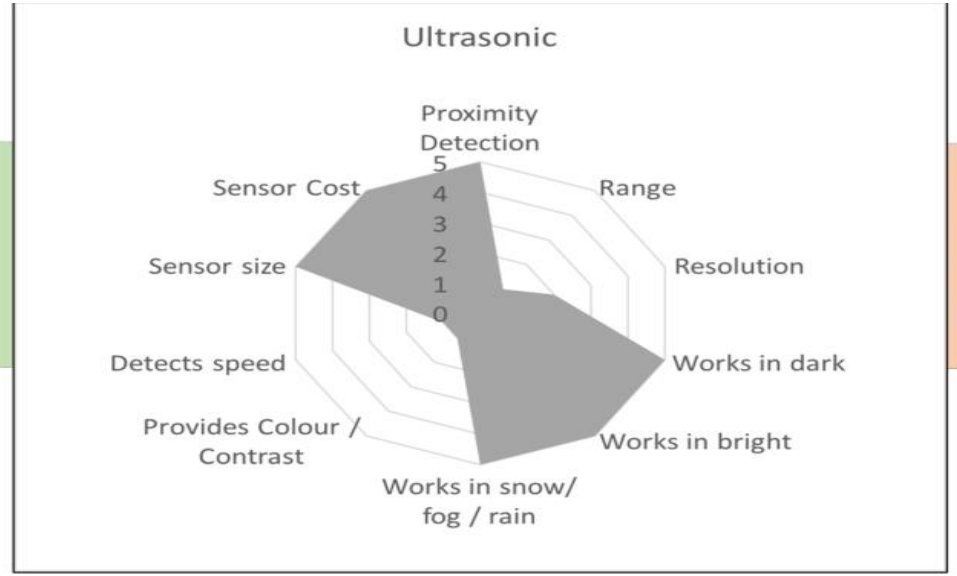
- Good detection in inclement weather, low-light condition; but at close proximity only (< 5m)
- Vehicles in opposite direction are not reported

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ADAS Sensors - Ultrasonic

Pros:

- **Proximity detection**
- **All-light conditions**
- **All-weather conditions**



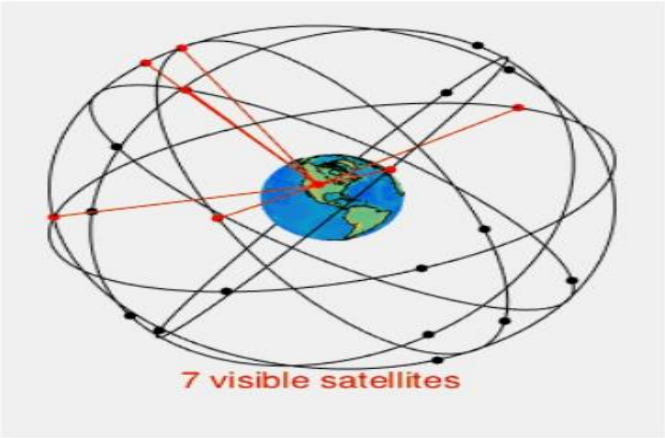
Cons:

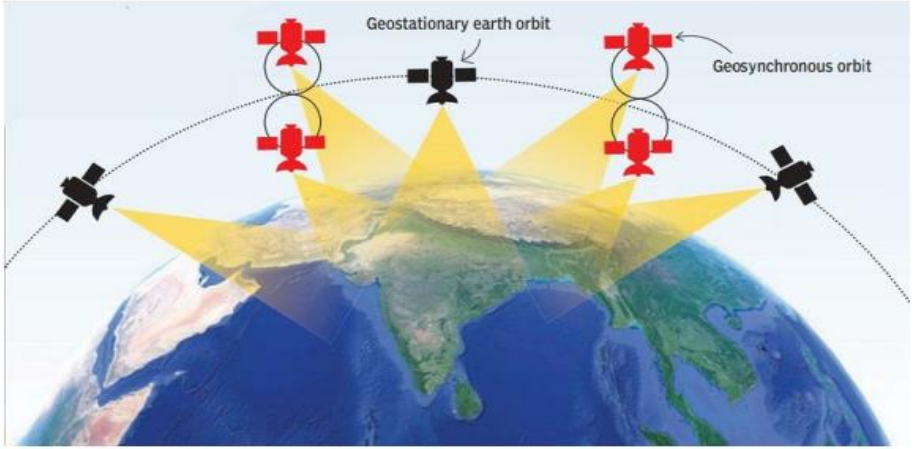
- **Range**
- **Speed detection**
- **Colour contrast**

Source: https://cleantechnica.com/2016/07/29/tesla-google-disagree-lidar-right/

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ADAS Sensors - GNSS

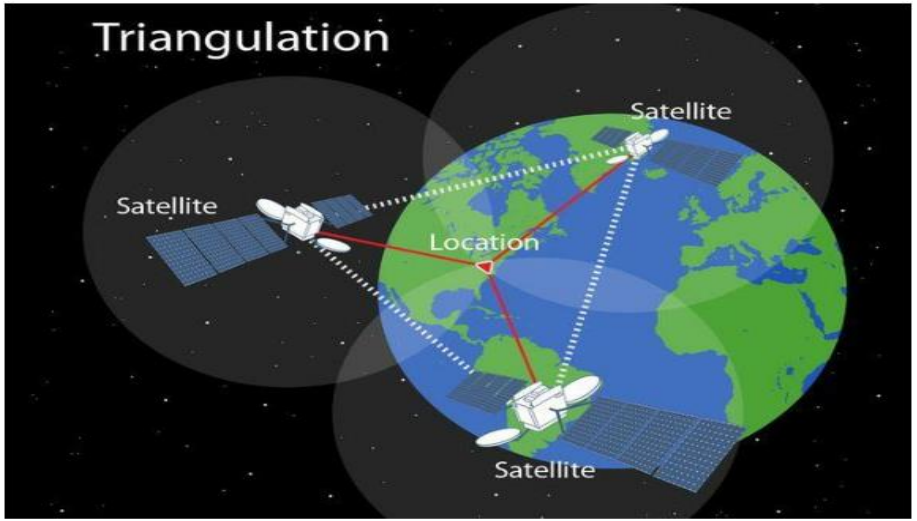




- GPS – US; GLONASS – Russia; GALILEO – Europe; Beidou – China; NAVIC – India; QZSS – Japan
  - GPS, GLONASS & GALILEO are global
  - Beidou, NAVIC & QZSS are regional
- GPS has 24 operating satellites, in 6 orbital planes covering the globe at ~20000 km altitude
- NAVIC has 7 satellites, 3 in geostationary and 4 in geosynchronous orbits covering the Indian subcontinent

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ADAS Sensors - GNSS



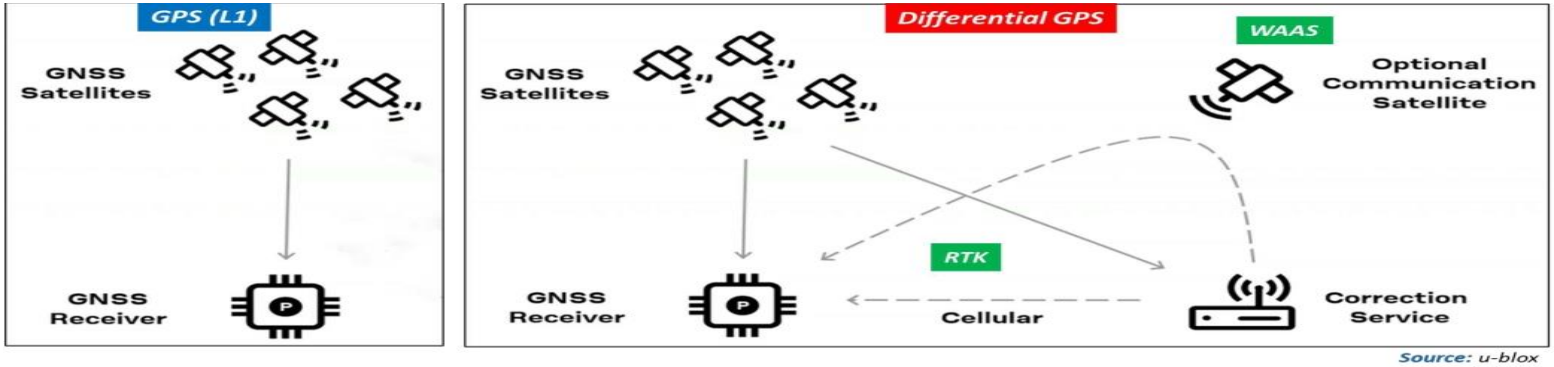
1. Satellites transmit their unique code along with position and transmit time information in the carrier frequency band
2. GPS receiver captures each satellite information and measures the received time
3. Delta time between transmit & receive is calculated; multiplied by the speed of light provides the range of that satellite
4. Ranges from three satellites form spheres that overlap but intersect only at one location. That will be the receiver's position

Source: National Geographic Society

- At any location on earth, and any time instance, at least four GPS satellites are in vicinity. Trilateration between three satellite positions help determine the location on earth. High-definition (HD) maps covering topological features are used to indicate location in context of the surroundings

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ADAS Sensors - GNSS



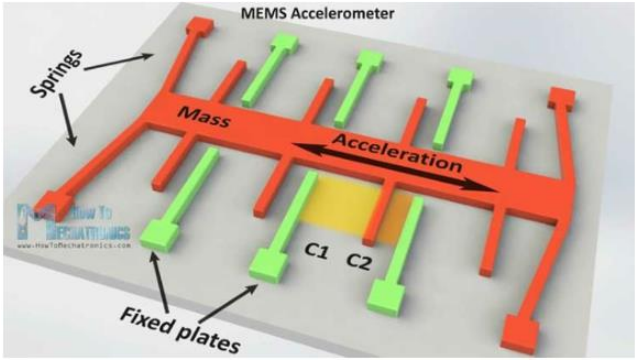
GPS (L1) precision is ~3m which is good for navigation; however for precise positioning this is not sufficient

Precision is improved by fixed & calibrated base-stations that:

- ✦Estimates the errors given its known location and calculated value
- ✦Broadcasts correction data over cellular or satellite radio for all other GPS receivers
- ✦Relies on the fact that the errors seen by two observers (near vicinity) on earth from a satellite constellation are same

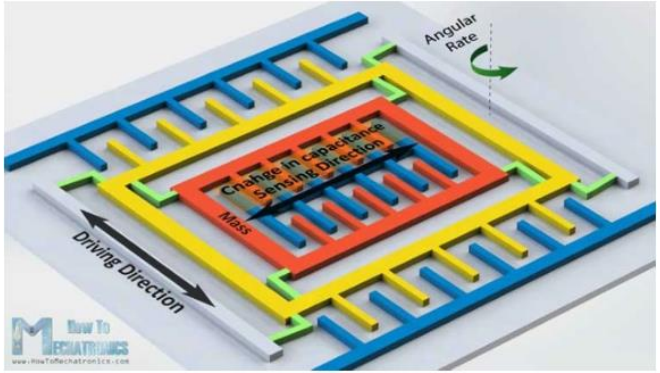
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ADAS Sensor – IMU



Source: https://howtomechatronics.com/

- Measures linear acceleration (m/s2)
- Spring mass moves forward or back, in the direction of acceleration or deceleration
- As spacing between fixed & moving plates change, effective capacitance C1 & C2 also vary
- Signal applied at the fixed plates and measured on moving plates; using this, acceleration can be calculated

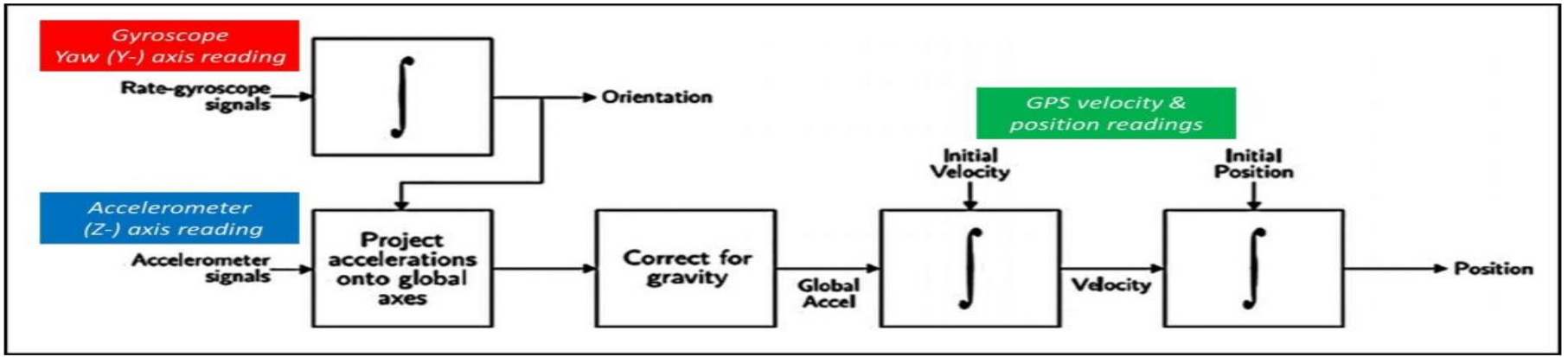


Source: https://howtomechatronics.com/

- Measures angular velocity (°/s (or) rad/s)
- Spring mass move perpendicular to the driving direction As spacing between fixed & moving plates change, effective capacitance C1 & C2 also vary
- Signal applied at the fixed plates and measured on moving plates; using this, angular momentum can be calculated

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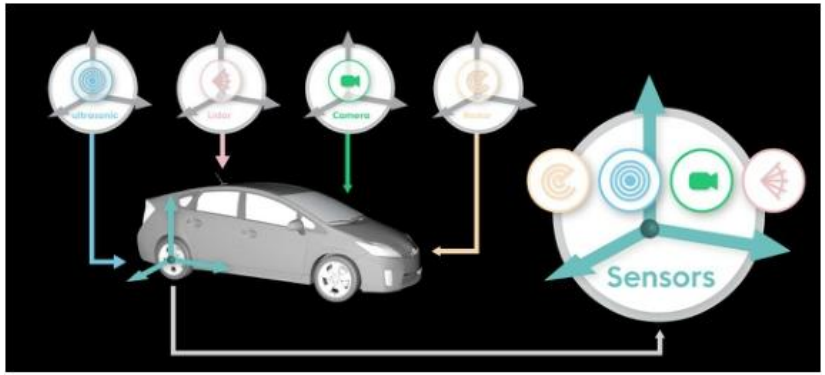
ADAS Sensor – IMU

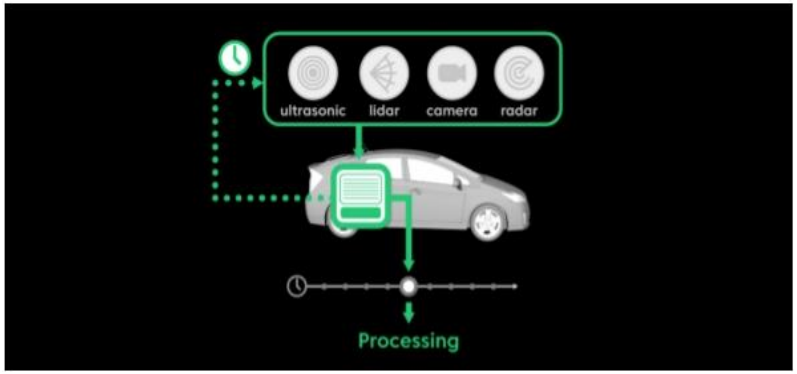


- Continuous heading and position data is updated using GPS, IMU and wheel tick measurements
- In the absence of GPS (in urban canyons & tunnels), IMU takes last known data and keeps updating on it
- Once GPS is re-established, last position of IMU data is provided as initial estimate to quickly converge

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ADAS Sensor – Sensor Fusion





- **Spatial Alignment** ***Source:*** *AIMotive*
  - All sensors are with reference to a common co-ordinate frame to compare or merge raw data
  - Usually factory calibrated, but requires periodic & automated calibration in field
  - Camera orientation is crucial to not distort the data and lead to erroneous detection (Ego motion)
- **Temporal Synchronization**
  - Camera, Lidar and Radar can provide data at different frame-rates using local clocks
  - Synchronization of all local clocks with the main processing clock is needed; raw data is always time coded
- **Reconstructing real world map**
  - Identify & track objects in real time with data fused from multiple sensors (Use Kalman filters to compute & update in real-time)

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Radar

- Robust and mature technology
- Types
  - SRR (<50m)
  - MRR (up to 150m)
  - LRR (up to 250m)



- Reliable operation in most weather conditions
- Accurate measurements of object velocity



- Applications
  - Adaptive cruise control (ACC)
  - Blind spot detection (BSD)
  - Forward collision warning (FCW)

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Radar

RADAR(Radio Detection and Ranging)

- The RADAR systems for intelligence vehicle work at frequencies of 24/77/79GHz,known as millimeter wave radar(MMW)
- Its principle is based on the measurement of emitter and the object by calculating the time of flight of the emitted signal and received echo
- Gives direction and speed of the Targets
- Use array of antennas capable of gathering a set of lobes that allows improvement of the range and a processing system for the detection of multiple targets





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ADAS Systems & Features

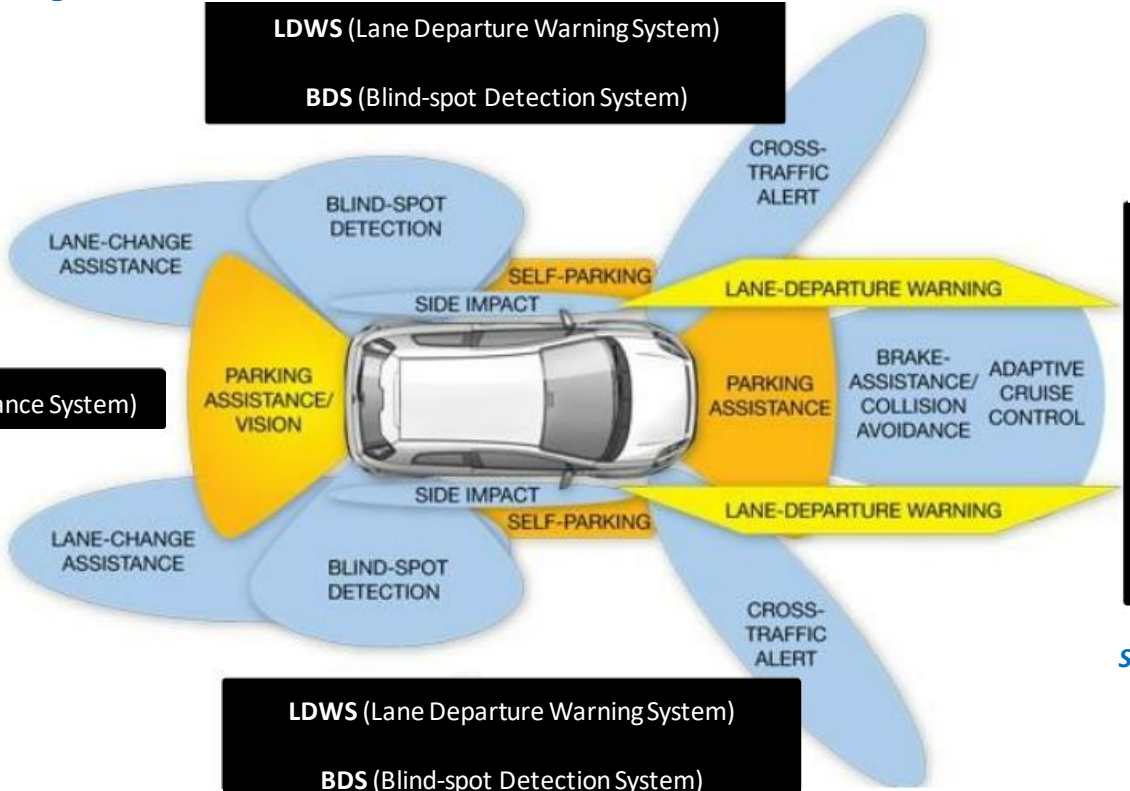
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ADAS Systems

LDWS (Lane Departure Warning System)

BDS (Blind-spot Detection System)

PAS (Parking Assistance System)



LDWS (Lane Departure Warning System)

BDS (Blind-spot Detection System)

ACC (Adaptive Cruise Control)

FCW (Forward Collision Warning)

AEBS (Automatic Emergency Braking System)

TSRS (Traffic Sign Recognition System)

Night Vision

Pedestrian Detection

Source: Ansys

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ADAS Systems – Adaptive Cruise Control (ACC)

Adaptive Cruise Control

- Consists of a front-mounted Radar sensor, Digital signal processor, and longitudinal controller
- Based on sensor information, instruction to the vehicle controller to slow down for any vehicle detected ahead
- Assists drivers to maintain a safe (pre-programmed or based on speed) separation distance or constant speed between vehicle
- Integrated with the brake and acceleration system, so that it automatically decelerates the vehicle and allow the vehicle to reaccelerate to the set speed when the path is clear
- The sensor will have limitations to detect:
  - *Stationary and slow moving vehicles*
  - *Pedestrians or objects in the roadway*
  - *Oncoming vehicles in the same lane*





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Environmental Perception: RADAR

RADAR

- Blind Spot Detection (BSD)
- Lane Change Assistant (LCA)
- Rear Cross Traffic Alert (RCTA)
- Forward Cross Traffic Alert (FCTA) or radar video fusion
- Radar waves have higher penetrability because they offer good features in all weather conditions, and can accurately detect short-range targets in front, to the side, and to the rear side of vehicle
- RADAR can significantly improve vehicle safety performance and reduce the decision-making burden of people at the wheel
- It can be installed besides the bumpers of the vehicle

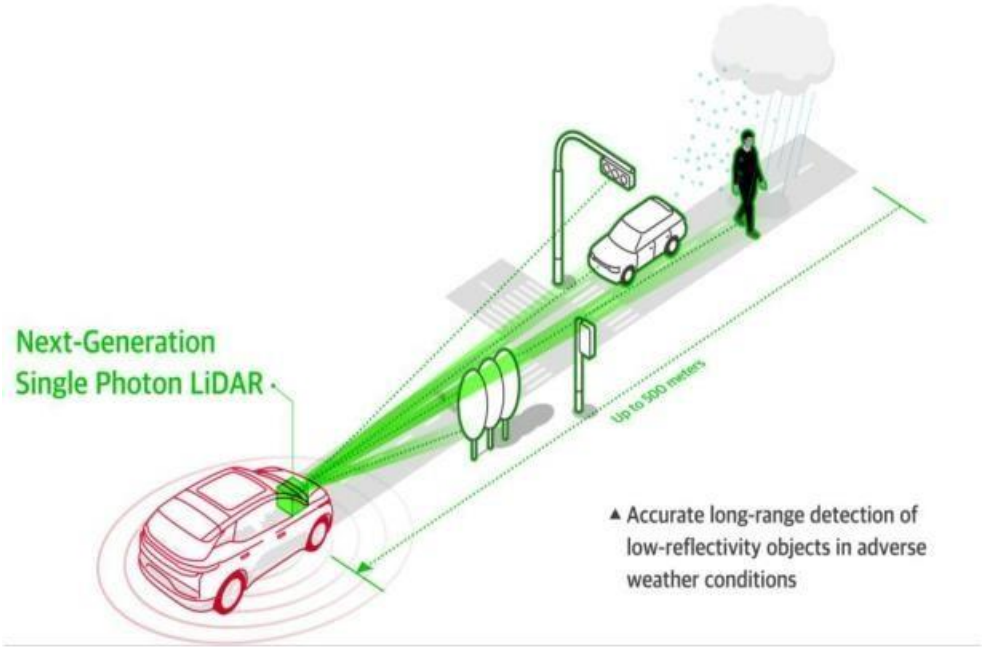


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Environmental Perception : LIDAR

Lidar (Light detection and Ranging)

- LiDAR's measures the Time of flight of a pulsed light emitted from a laser diode until it is received by an emitter
- Types
  - 2 D LiDAR
  - 3 D LiDAR
  - Solid state LiDAR



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LIDAR Challenges

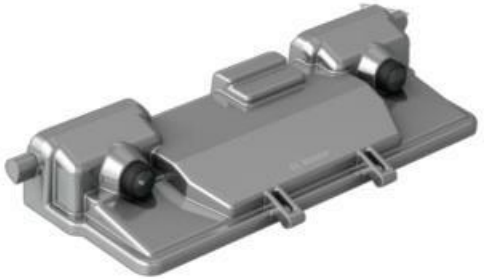
- Cost
- Miniaturization for easy integration in car body
- High aperture angle – number of channels
- Mechanical and electronics reliability
- High Data transfer rates (> 1 GB per minute depending upon number of channels)
- Low refresh rates ~ 10 to 30Hz
- Noisy in environments with suspended particles, dust, rain and fog
- Lot of new offerings and players



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Camera

- Fast image acquisition
- Stereo camera for depth and distance of the object in image plane
- High resolution AND high speed
- Resolution: greater than 1k x 1k pixels, aim for 4k x 4k
- Frame rate: 60 fps and higher for shorter reaction time
- Zoom into image area for a greater detail
- Efficient image processing for real-time analysis
- Wider field of view than RADAR or LiDAR



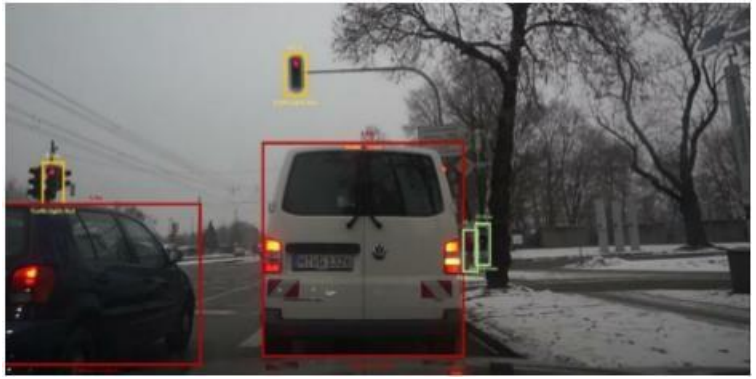


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Camera

Detection of environment:

- Object detection and classification
- Object dynamics and direction of motion
- Lane detection (Real and virtual)
- Traffic Sign Detection
- Traffic light classification
- Curb Detection
- Detection of speed bumps
- Free space detection
- Localization in fusion with GNSS/IMU





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Position Estimation Systems

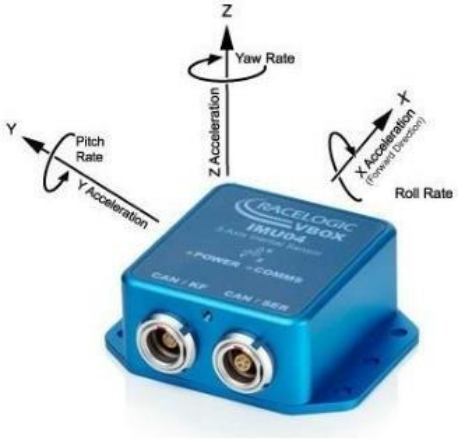
- Positioning systems for vehicles are hybrid systems that merge data from different sources such as odometry, IMUs, GNSS ,LiDAR's, RADARs and cameras to obtain a reliable position
- GNSS:-
  - The GNSSs provides the absolute position of a receiver with respect to a fixed reference and consists of a set of satellites orbiting approximately 20,000 km from the earth's surface
  - The operating principal of the GNSS is based on measuring the time of flight of the signal emitted by the satellite and that received by the receptor. The system is able to reliably obtain details of position and time (x, y, z, t) with a minimum of four visible satellites.



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Inertial Measurement Unit

- Inertial Measurement Unit (IMU) is a sensor capable of defining the movement of the vehicle along the yaw, pitch, roll axis. This sensor calculates acceleration along the X, Y, Z axes, orientation, inclination, and altitude
- IMU consists of three types of motion sensors
  - Accelerometer
  - Gyroscope
  - Magnetometer



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IMU- Mounting Options

1. On the roof







- With its three points of contact the Vacuum Tripod Mount (RLACS256) ensures a very stable fixture and reduces IMU vibration that might be transmitted by the vehicle's roof.
- The adjustable lever arms allow for the levelling of the IMU on a non-flat roof whilst extending the mounting options on the vehicle.

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IMU- Mounting Options

2. Within the Vehicle using a Mounting Arm

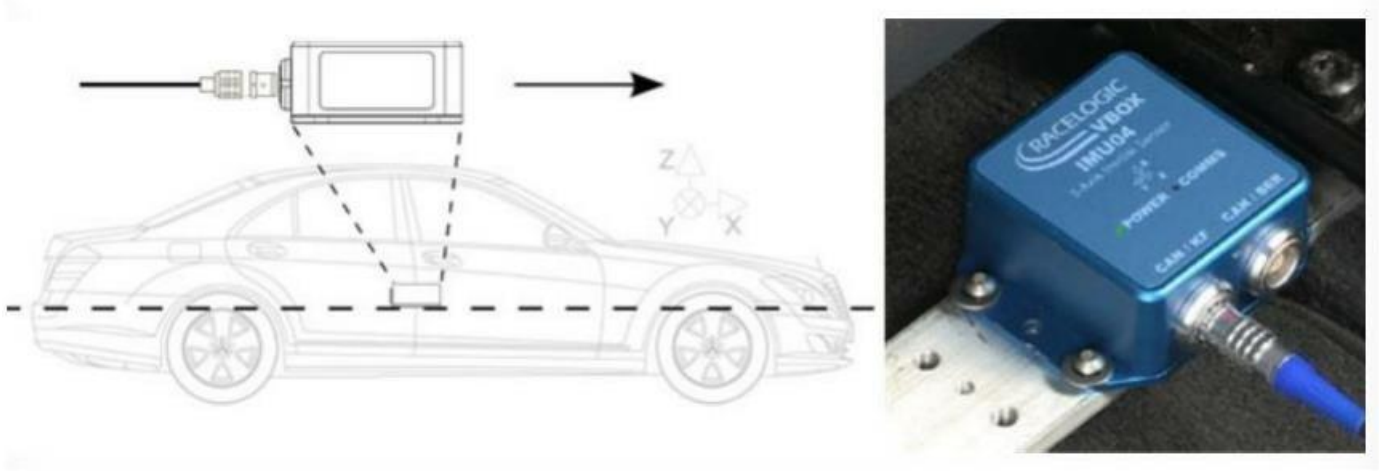


- If roof mounting the IMU is not possible the IMU can be fixed inside the vehicle. A flexible way to fix the IMU rigidly within the vehicle is by using the mounting arm.
- Pressed against the IMU on the floor and the vehicle's ceiling, the mounting prop ensures that the IMU is fixed tightly.

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IMU- Mounting Options

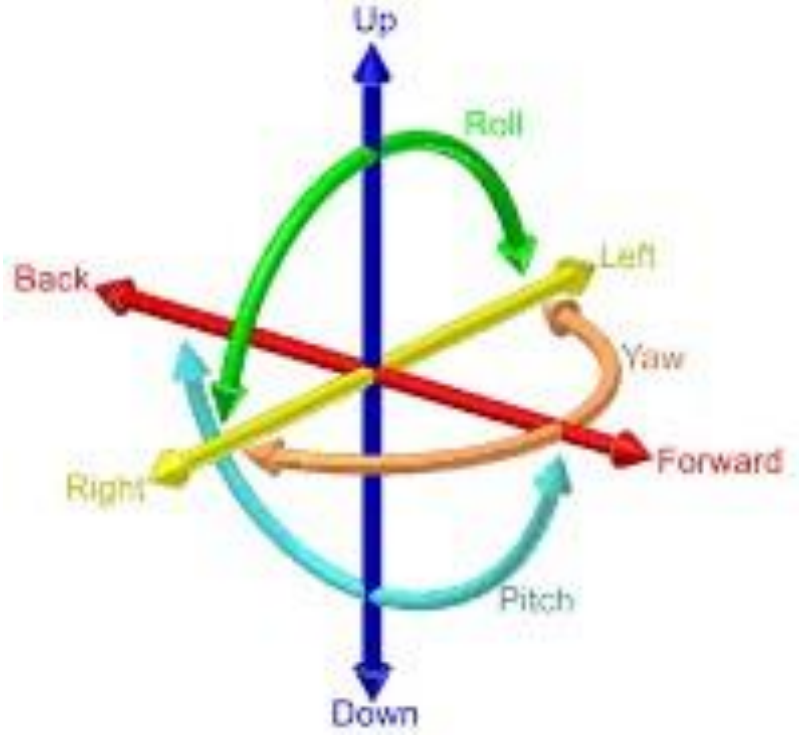
3. Within the Vehicle - Fixed to vehicle body



- Make sure it is mounted in the direction of travel.
- Mount the IMU and GPS antenna as close to each other as possible.

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VBOX IMU (Inertial Measurement Unit)







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Inertial Measurement Unit



[High accuracy measurements of Pitch, Roll and Yaw using an Inertial Measurement Unit - YouTube]

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HD Maps

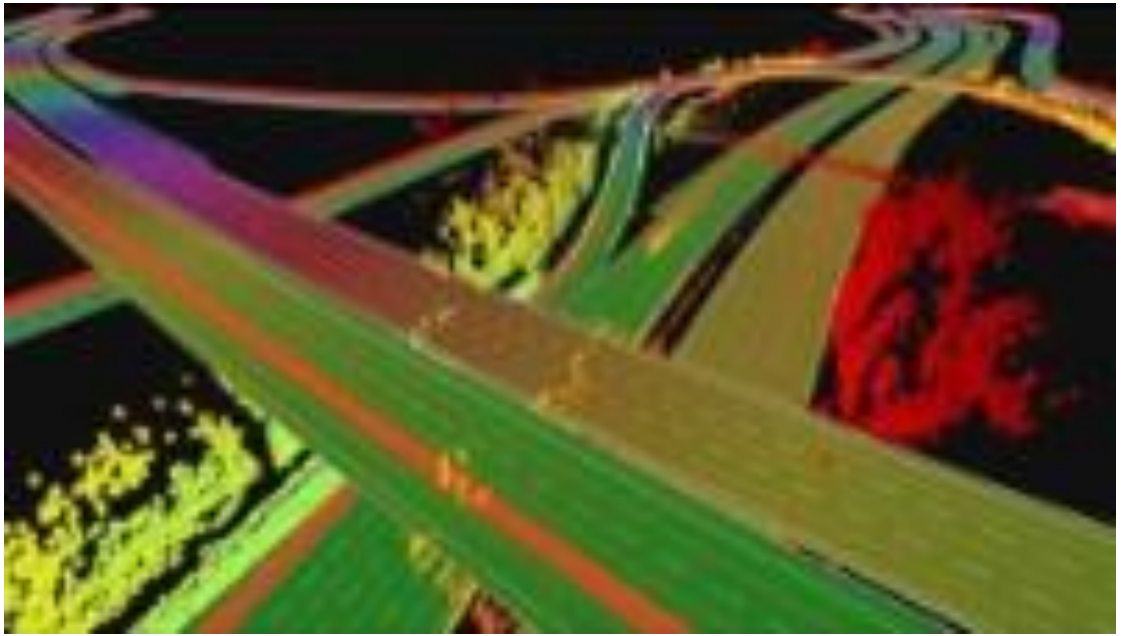
- Detailed inventory
  - Lanes
  - Road edges
  - Shoulders
  - Dividers
  - Traffic signals
  - Traffic signs
  - Paint marking
- Predictive awareness so that vehicles know what lies ahead





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Tom Tom



https://youtu.be/ga5fW-QSXp0

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HERE Technologies

Founded in 1985
Located in Amsterdam (Netherlands) and Berkeley (US)

- Provides cloud-based navigation and mapping solutions for multiple industries such as automotive, transportation & logistics, public sector, retail and more.
- Offers a cloud-enabled platform to create, develop location-centric data assets and offers
- Enables users to create custom maps.
- Also offers Here marketplace which is an open-source platform to exchange & monetize location assets from trusted sources.

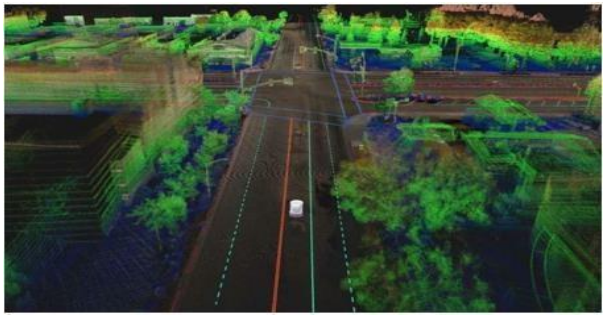


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DeepMap and NVIDIA

Founded in 2016 Located in Palo Alto (US)

- DeepMap Inc. develops software tools to enable autonomous cars to create centimeter-level accurate real-time maps of their surroundings.
- It uses data from LIDAR and optical cameras to build its maps and localizes vehicles within these real- time maps.
- Its intended business model is to license the software to traditional car makers and tech startups in the autonomous car space
- DeepMap and NVIDIA have gained in increasing the global presence and enhancing the performance of HD maps



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ADAS Systems – Adaptive Cruise Control (ACC)



- Matlab / Simulink (PAS toolbox) supports multiple scenarios to simulate various corner conditions for ACC

Source: https://www.youtube.com/watch?v=own_VaRZ9M8

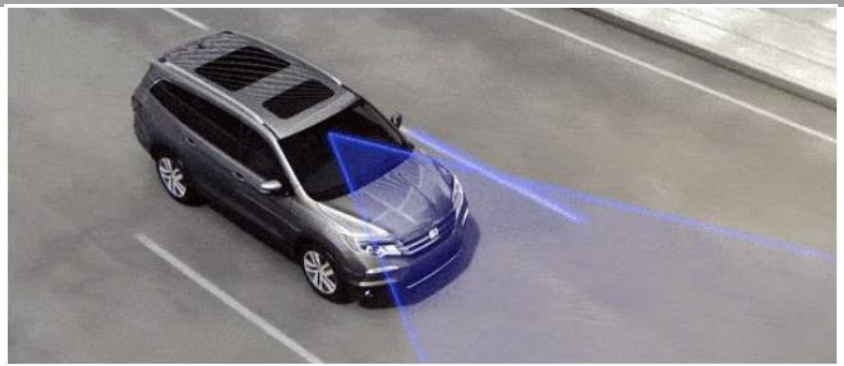
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ADAS Systems – Lane Departure Warning (LDW)

Lane Departure Warning

- Uses a wide-angle camera mounted front of the rearview mirror to recognize striped and solid lane markings
- Generates audio-visual alerts if the vehicle starts deviating from its lane without an appropriate turn signals
- Extending the departure warning system, the Lane Keeping Assistance (LKA) helps to
  - *Counter-steer gently through the electronic power steering*





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ADAS Systems – Lane Change Assist (LCA)

Lane Change Assist

- Uses two (or four) medium-range (MR) Radar sensors mounted on the corners of the vehicle
- The radar data stream provides complete picture of vehicles ahead or approaching with speed information
- When another vehicle is approaching from behind, or already present in the blind spot
  - A warning light in the side mirror alerts the driver to the hazard
  - Additional acoustic / haptic warnings should the driver activate the turn indicator



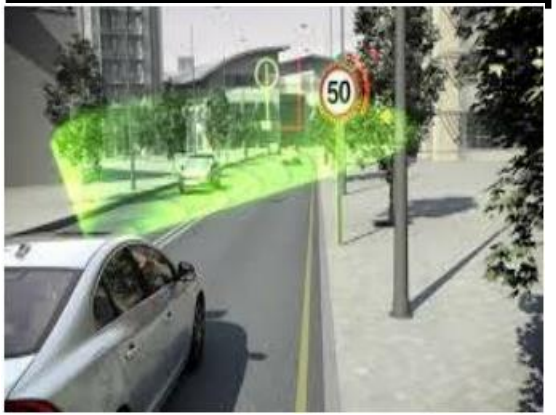
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ADAS Systems – Traffic Sign Recognition (TSR)

Traffic Sign Recognition System

- Forward-facing wide-angle lens Cameras to capture road signs
- Image processing & image recognition algorithms (traditional or Machine-learning based) help recognize the traffic signs and display on the infotainment system
- Alerts distracted drivers to follow traffic rules, helping avoid accidents and fines





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ADAS Systems – Parking Assistance (PAS) & Night Vision



Parking Assistance (PAS)

- Uses ultrasonic sensors fixed on front and rear bumpers of the vehicle to detect the obstacles while parking and triggering audio alarm
- Rear camera is fused with Ultrasonic data to provide additional visual assistance



Night Vision and Pedestrian Detection

- Uses infrared sensors or stereoscopic cameras to capture images in low-visibility
- Infrared sensors capture thermal radiations from the front of the vehicle and clearly distinguish living beings from objects, trees and traffic signs on pedestrian and roadways

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ADAS Systems – Forward Collision Warning (FCW)

Forward Collision Warning

- Uses Radar/Lidar and Camera sensors to issue audio, visual and haptic alerts to collision
- System measures the distance, angular direction and relative speed between the two vehicles
- Works in conjunction with Adaptive Cruise Control (ACC), and can be integrated with Automatic Emergency Braking (AEB) systems to prevent collisions



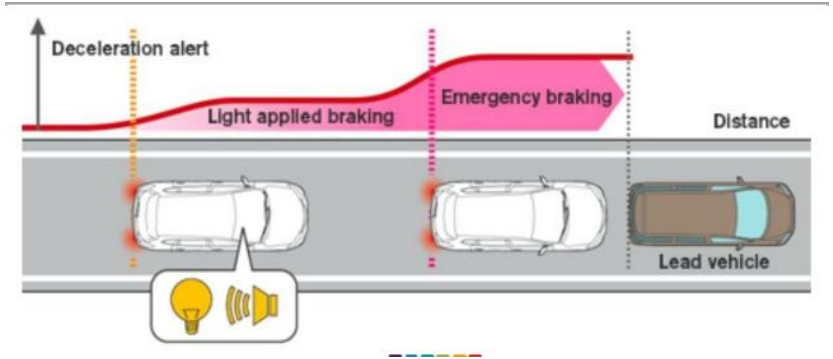


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ADAS Systems – Automatic Emergency Braking (AEB)

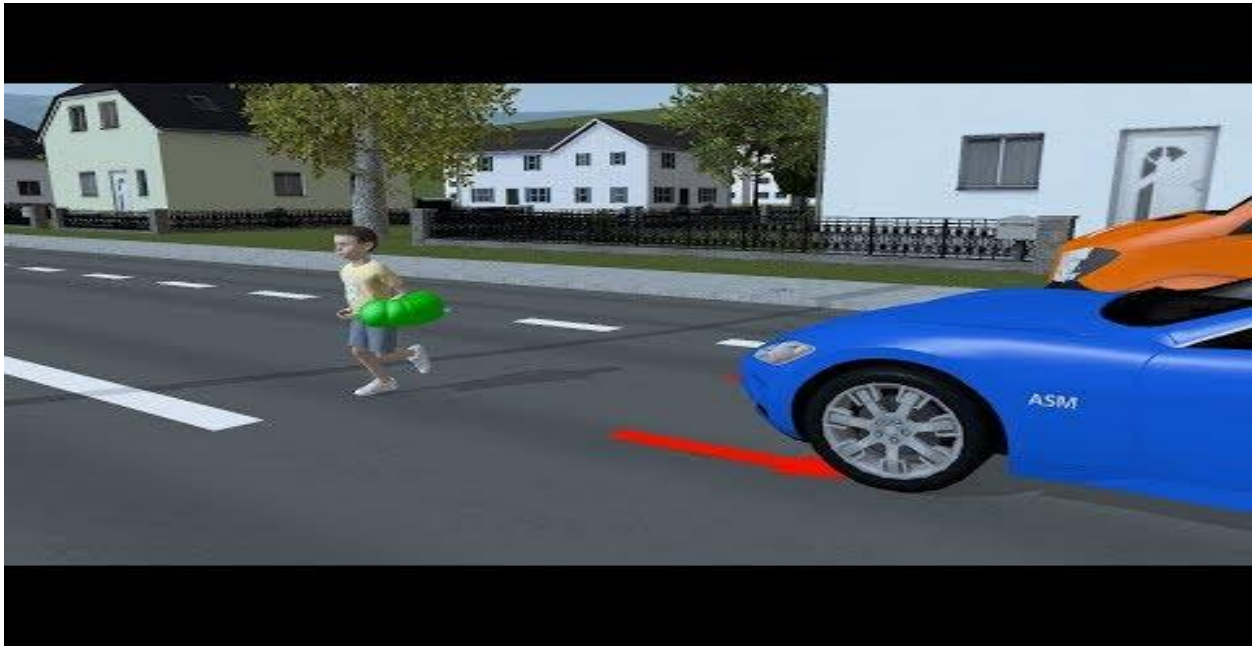
Automatic Emergency Braking (AEB)

- Uses Radar and Ultrasonic sensors to detect vehicle speed and clearance to objects, calculating required deceleration
- Prepares the braking system for potential emergency braking
- Provides Audible / Visual warning
- Short brake jerk given with enough time to react
- As soon as the driver presses the brake pedal, the system provides braking support
- If the driver fails to react to the immediate risk of collision, the automatic emergency braking engages

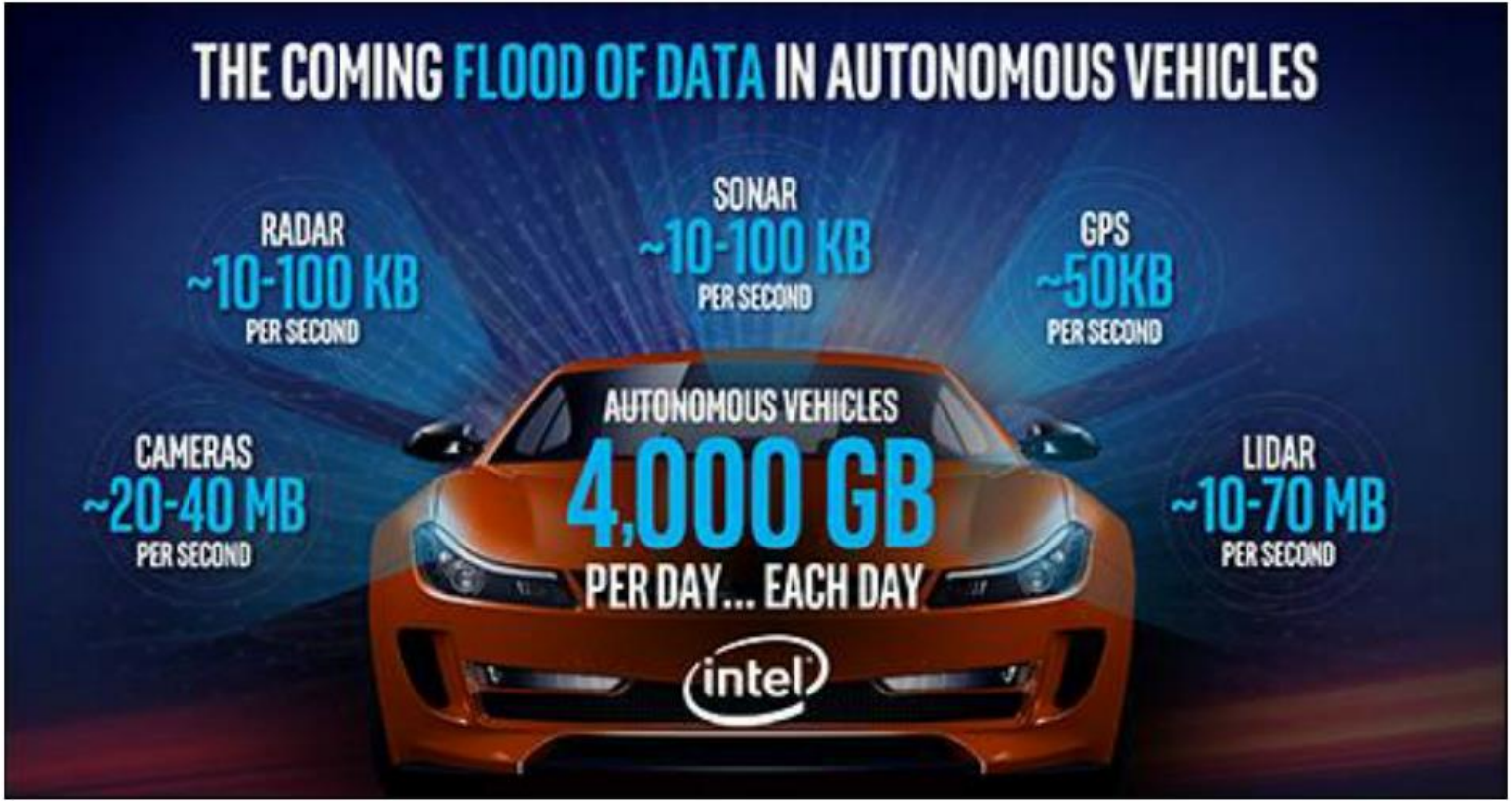


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ADAS Systems – Automatic Emergency Braking (AEB)



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http://newsletter.motor.com/2016/20161205/IID_DataIsTheNewOil.html

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ADAS Systems – Waymo Perception & Navigation



https://www.youtube.com/watch?v=2CVInKMz9cA

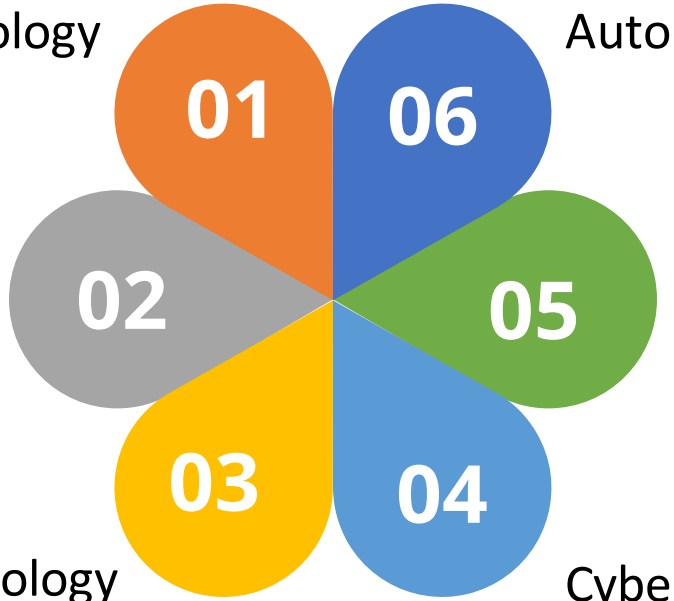
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Our Expertise

Electric Vehicle Technology

Future Mobility Overview

ADAS & Autonomous Technology



Automotive Electronics

IoT, Connectivity & Advanced Computing

Cyber Security for Automotive

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